Plasma Etching System

Equipment Technical Documentation (Sample)

---

1. Equipment Overview

The Plasma Etching system is used to selectively remove material from semiconductor wafers using reactive plasma. The process involves ionized gases that chemically or physically etch thin films.

Tool Type: Plasma Etch
Model: ETCH-Z200
Communication Protocol: SECS/GEM (HSMS)

---

2. Process Parameters (Variables)

2.1 Status Variables (SV)

VID	Name	Type	Unit	Description
V1101	ChamberTemperature	Float	°C	Chamber temperature
V1102	ChamberPressure	Float	mTorr	Chamber pressure
V1103	RFPower	Float	W	RF power applied to plasma
V1104	BiasVoltage	Float	V	Bias voltage applied
V1105	GasFlowRate_CF4	Float	sccm	CF4 gas flow rate
V1106	ProcessStatus	String	-	Current system status

2.2 Data Variables (DV)

VID	Name	Type	Unit	Description
V2101	RecipeID	String	-	Active etching recipe
V2102	WaferID	String	-	Current wafer identifier
V2103	EtchTime	Float	sec	Duration of etching process
V2104	EtchDepth	Float	nm	Depth of material removed

---

3. Equipment Events (CEID)

CEID	Event Name	Description	Linked VIDs
E201	ProcessStart	Etching process initiated	V1101, V1102, V2101
E202	ProcessEnd	Etching process completed	V2103, V2104
E203	PlasmaIgnition	Plasma successfully ignited	V1103, V1104
E204	WaferLoaded	Wafer loaded into chamber	V2102
E205	WaferUnloaded	Wafer removed from chamber	V2102
E206	AlarmTriggered	Alarm condition detected	V1101, V1102, V1103

---

4. Alarm Definitions

Alarm ID	Name	Severity	Description
A301	OverTemperature	Critical	Chamber temperature exceeds limit
A302	PressureInstability	Warning	Pressure fluctuation detected
A303	PlasmaFailure	Critical	Plasma ignition failed
A304	GasFlowMismatch	Warning	Gas flow rate deviation

---

5. Remote Commands (RCMD)

Command	Description	Parameters
START	Start etching process	RecipeID
STOP	Stop process	None
PAUSE	Pause etching	None
RESUME	Resume etching	None
RESET	Reset system after error	None

---

6. Equipment States

State ID	State Name	Description
S1	Idle	System ready

State ID	State Name	Description
S2	Processing	Etching in progress
S3	Completed	Process finished
S4	Error	Fault condition
S5	Maintenance	System under maintenance

---

7. State Transitions

- Idle → Processing (Triggered by START command)
- Processing → Completed (Triggered by ProcessEnd event)
- Processing → Error (Triggered by PlasmaFailure alarm)
- Error → Idle (Triggered by RESET command)
- Idle → Maintenance (Manual transition)

---

8. Communication Details

- Protocol: SECS/GEM
- Transport: HSMS (TCP/IP)
- Port: 5001

Example Messages:

- S1F1 → Communication request
- S6F11 → Event notification (ProcessStart, PlasmaIgnition)
- S2F41 → Remote command execution (START, STOP)

---

9. Notes

- Plasma stability is critical for uniform etching
- Gas flow calibration must be verified before process start
- All alarms must be acknowledged before system reset